PROGRAM:

/* program for bubble sort */

```
#include<stdio.h>
void bubbleSort(int array[],int n)
{
      int i,j;
  for (i=0;i<n-1;++i)
{
      for(j=0;j<n-i-1;++j)
      {
    if(array[j]>array[j+1])
    {
     int temp=array[j];
        array[j]=array[j+1];
        array[j+1]=temp;
    }
   }
   printArray(array,n);
}

}

void printArray(int array[],int n)
{
      int i;
      for(i=0;i<n;++i)
{
      printf("%d ",array[i]);
}
   printf("\n");
}
 int main()
{
      int data[]={5,4,8,1,2};
      int n=sizeof(data)/sizeof(data[0]);
      printf("unsorted array:\n");
      printArray(data,n);
      bubbleSort(data,n);
      printf("sorted Array ascending order:\n");
      printArray(data,n);
}
```

OUTPUT:

```
unsorted array:
5 4 8 1 2
4 5 1 2 8
4 1 2 5 8
1 2 4 5 8
1 2 4 5 8
sorted Array ascending order:
1 2 4 5 8
```